MID-TERM EXAMINATION

(Course Name: B. Tech.) (Semester: Ist)

(October, 2025) OFF LINE mode

| Subject Code: BAS 104 | Subject: Environmental Sciences |
|---|---|
| Time : 1 ½Hours | Maximum Marks :30 |

**Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3**

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| **Q1** | | **(2.5*4=10)** | **CO Mapping** |
| | **a)** Calculate the temporary and total hardness of a water sample containing: $Mg(HCO_3)_2$ = 73 ppm, $Ca(HCO_3)_2$= 162 ppm, $MgCl_2$ = 95 ppm | | **CO1** |
| | **b)** What are the major approaches to conserve water resources? How is water harvesting practiced in India? | | **CO1** |
| | **c)** Discuss the sources and sinks of nitrogen oxides as air pollutants. | | **CO2** |
| | **d)** Calculate atom economy of production of Chlorobenzene in the following reaction: $C_6H_6 + Cl_2 \longrightarrow C_6H_5Cl + HCl$ | | **CO2** |
| | **UNIT I** | | **CO Mapping** |
| **Q2** | **Attempt any two parts** | **(5*2=10)** | **CO1** |
| | **a)** 100 ml of water sample on titration with N/50 HCl requires 8 mL of acid to phenolphthalein end point and 9 mL of acid to methyl orange end point. Calculate the type and extent of alkalinity present in water sample. | | |
| | **b)** How is water treated to make it suitable for human consumption? Illustrate the significance of break-point chlorination. | | |
| | **c)** Explain various tertiary treatment techniques used to improve the quality of waste water. | | |
| | **UNIT II** | | **CO Mapping** |
| **Q3** | **Attempt any two parts** | **(5*2=10)** | **CO2** |
| | **a)** What devices/methods are being used by industries to control particulate emissions? Discuss any two briefly. | | |
| | **b)** Write short notes on:<br>(i) Selection of renewable feedstocks as starting material in a process.<br>(ii) Artificial $CO_2$ Sequestration. | | |
| | **c)** Describe various treatment and disposal methodologies being used to manage solid waste. | | |